Integration by Substitution

Section 1: Basic u-Substitution

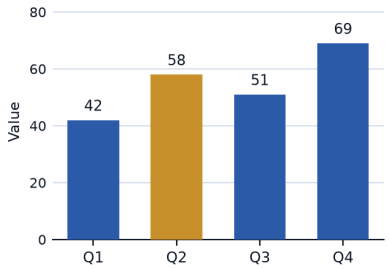
1 Evaluate $\int 2x(x^2+5)^4\,dx$.

a. $\frac{1}{5}(x^2+5)^5+C$

b. $\frac{1}{4}(x^2+5)^4+C$

c. $(x^2+5)^5+C$

d. $\frac{2}{5}(x^2+5)^5+C$



2 Evaluate $\int x^2\sqrt{x^3+1}\,dx$.

a. $\frac{2}{9}(x^3+1)^{3/2}+C$

b. $\frac{2}{3}(x^3+1)^{3/2}+C$

c. $\frac{1}{3}(x^3+1)^{3/2}+C$

d. $\frac{3}{2}(x^3+1)^{1/2}+C$

3 Evaluate $\int \frac{x}{x^2+9}\,dx$.

a. $\frac{1}{2}\ln(x^2+9)+C$

b. $\ln(x^2+9)+C$

c. $\frac{1}{3}\arctan(x/3)+C$

d. $\frac{1}{2(x^2+9)^2}+C$

Section 2: Trigonometric and Exponential Substitutions

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

4 Evaluate $\int \sin^3(x)\cos(x)\,dx$.

a. $\frac{1}{4}\sin^4(x) + C$

b. $\frac{1}{3}\sin^3(x) + C$

c. $-\frac{1}{4}\cos^4(x) + C$

d. $\sin^4(x) + C$

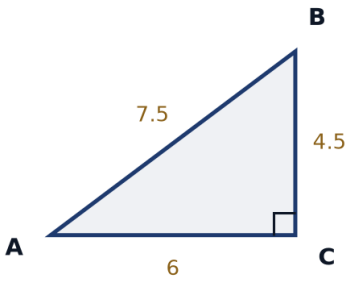
5 Evaluate $\int \tan(x)\,dx$.

a. $\ln|\sec x| + C$

b. $-\ln|\sec x| + C$

c. $\sec^2(x) + C$

d. $\ln|\sin x| + C$



6 Evaluate $\int e^{3x-1}\,dx$.

a. $\frac{1}{3}e^{3x-1} + C$

b. $3e^{3x-1} + C$

c. $e^{3x-1} + C$

d. $\frac{1}{3}e^{3x} + C$

Section 3: Definite Integrals with Limits Change

7 Evaluate $\int_0^1 x(x^2+1)^3\,dx$.

a. 15/8

b. $16/8 = 2$

c. 7/8

d. 15/4

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

8 Evaluate $\int_0^{\sqrt{\pi}} x\sin(x^2)\,dx$.

a. 1

b. 0

c. 2

d. 1/2

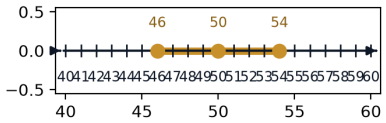
9 Evaluate $\int_1^e \frac{\ln x}{x}\,dx$.

a. 1/2

b. 1

c. 2

d. $1/e$



Section 4: Advanced Linear Substitutions

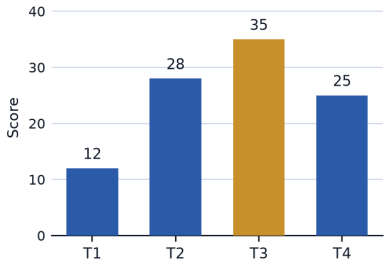
10 Evaluate $\int x\sqrt{x+2}\,dx$.

a. $\frac{2}{5}(x+2)^{5/2} - \frac{4}{3}(x+2)^{3/2} + C$

b. $\frac{2}{3}(x+2)^{3/2} + C$

c. $\frac{1}{5}(x+2)^{5/2} - \frac{2}{3}(x+2)^{3/2} + C$

d. $\frac{2}{5}x^{5/2} + C$



11 Evaluate $\int \frac{x}{\sqrt{x+1}}\,dx$.

a. $\frac{2}{3}(x+1)^{3/2} - 2\sqrt{x+1} + C$

b. $\frac{1}{3}(x+1)^{3/2} - \sqrt{x+1} + C$

c. $2\sqrt{x+1} + C$

d. $\frac{2}{3}(x+1)^{3/2} + C$

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

12 Evaluate $\int \frac{e^{2x}}{e^x + 1}\, dx$.

a. $e^x - \ln(e^x + 1) + C$

b. $e^x + \ln(e^x + 1) + C$

c. $\ln(e^x + 1) + C$

d. $\frac{1}{2}e^{2x} + C$

© ClipSAT: By Mr. Mohamed Abdallah +966597688647